

JUNSEUNG YOU (유준승)

Seoul, South Korea | jsyou@sor.snu.ac.kr | 82 10 6652 5760 | sor.snu.ac.kr | github.com/junseungyou

SUMMARY

I am a postdoctoral researcher at Seoul National University (SNU) Institute of Computer Technology, currently serving my mandatory military service as expert research personnel. I earned my Ph.D. from Security Optimization Lab (SOR Lab) in the Department of Electrical and Computer Engineering at SNU, under the supervision of Professor Yunheung Paek. My research agenda develops **hardware-assisted mechanisms that efficiently address fundamental systems security challenges** such as memory safety and isolation, leveraging architectural primitives including Arm Memory Tagging Extension (MTE) and trusted execution environments. My work is grounded in low-level systems development across the compiler, operating system, and hypervisor.

RESEARCH INTERESTS

- Systems Security
- Hardware-assisted Security
- Confidential Computing
- Autonomous Systems Security

EDUCATION

Seoul National University , Ph.D. in Electrical and Computer Engineering	Sep 2019 - Feb 2026
Advisor: Yunheung Paek	
Dissertation Title: Hardware-Assisted Memory Protection on ARM with Memory Tagging Extension	
Seoul National University , B.E. in Electrical and Computer Engineering	Mar 2014 - Aug 2019
Korean Minjok Leadership Academy	Mar 2011 - Feb 2014

EXPERIENCE

Seoul National University , Seoul, Korea	Mar 2026 - Aug 2027
<i>Military Service</i>	
Serving as Expert Research Personnel (전문연구요원)	
Seoul National University , Seoul, Korea	Sep 2019 - Feb 2026
Research Assistant	
Advisor: Yunheung Paek	
Arizona State University , Tempe, AZ, USA	Jan 2024 - Feb 2024
Visiting Researcher	
Advisor: Gail-Joon Ahn, Hokeun Kim	
Topic: Hardening Arm trusted execution environment (confidential compute architecture)	
National University of Singapore , Singapore, Singapore	Sep 2018 - Feb 2019
Research Intern	
Advisor: Min Suk Kang	
Topic: Securing network intrusion detection system with trusted execution environment	

PUBLICATIONS

Refereed Conference Publications

- [1] **SECV: Securing Connected Vehicles with Hardware Trust Anchors** Feb 2026
Martin Kayondo*, **Junseung You***, Eunmin Kim, Jiwon Seo, and Yunheung Paek
(*: Both authors contributed equally to this work)
Network and Distributed System Security (NDSS) Symposium
- [2] **BASTAG: Byte-level Access Control on Shared Memory using ARM Memory Tagging Extension** Oct 2025
Junseung You, Jiwon Seo, Kyeongryong Lee, Yeongpil Cho, and Yunheung Paek
ACM SIGSAC Conference on Computer and Communications Security (CCS)

- [3] **KVSEV: A Secure In-Memory Key-Value Store with Secure Encrypted Virtualization** Oct 2023
Junseung You, Kyeongryong Lee, Hyungon Moon, Yeongpil Cho, and Yunheung Paek
ACM Symposium on Cloud Computing (SoCC)
- [4] **SFITAG: Efficient Software Fault Isolation with Memory Tagging for ARM Kernel Extensions** July 2023
Jiwon Seo, **Junseung You**, Yungi Cho, Yeongpil Cho, Donghyun Kwon, and Yunheung Paek
ACM Asia Conference on Computer and Communications Security (ASIACCS)

Refereed Journal Publications

- [5] **ZOMETAG: Zone-based Memory Tagging for Fast, Deterministic Detection of Spatial Memory Violations on ARM** July 2023
Jiwon Seo*, **Junseung You***, Donghyun Kwon, Yeongpil Cho, and Yunheung Paek
(*: Both authors contributed equally to this work)
IEEE Transactions on Information Forensics and Security (TIFS)
- [6] **Enhancing a Lock-and-Key Scheme with MTE to Mitigate Use-After-Frees** Dec 2023
Inyoung Bang, Martin Kayondo, **Junseung You**, Donghyun Kwon, Yeongpil Cho, and Yunheung Paek
IEEE Access
- [7] **SBGen: A Framework to Efficiently Supply Runtime Information for a Learning-based HIDS for Multiple Virtual Machines** Nov 2020
Jiwon Seo, Inyoung Bang, **Junseung You**, Yeongpil Cho, and Yunheung Paek
IEEE Access

PROJECTS INVOLVED

with fundings, Principal Investigator : Yunheung Paek

- On-device AI Protection** funded by MCST and KOCCA, RS-2025-02221620 Apr 2025 - Current
• Role: Designing on-device AI model protection protocol with TEEs.
- Side-channel Resistant SoC IP** funded by Samsung Electronics, South Korea Sep 2024 - Sep 2025
• Role: Designing mask/parity-based side-channel protection mechanism on RISC-V SoC.
- Preserving Privacy using Confidential Computing** funded by IITP, RS-2024-00438729 Jun 2024 - Current
• Role: Designing user-defined policy-based data flow analysis runtime for privacy preservation inside TEEs.
- Autonomous Vehicle Security Monitor** funded by MOTIE, RS-2024-00406121 Apr 2024 - Current
• Role: Design ECU protection framework with multiple hardware trust anchors.
- HW-centric Secure Data Box** funded by IITP, 2021-0-00528 Apr 2021 - Current
• Role: Designing and implementing secure protocol for using multiple privacy-preserving techniques.
- Data Tracking and Protection in Cloud Edge** funded by IITP Apr 2020 - Dec 2021
• Role: Designing lightweight attestation framework for TEEs.

TEACHING EXPERIENCE

Seoul National University (TA)

Topics on System Software (Data Security and Privacy)

- Period : 2024-2, 2025-2
- Role : Head teaching assistant. Authored lecture materials for system security and TEEs. Lecturer for several system security related topics. Authored, administered, and graded presentations and lab projects.

Introduction to Security, Privacy and Blockchain

- Period : 2024-1, 2025-1
- Role: Authored course materials for system security and confidential computing. Lecturer for introduction to computer systems and security. Authored, administered, and graded course quizzes.

Creative Design Project (Graduation Project)

- Period : 2022-1 to Current
- Role : Guided undergraduates on their final-year graduation projects from topic selection to thesis writeup.

HONORS & AWARDS

Postdoctoral research funding <i>SNU's BK21 FOUR program for Future Leaders in Information Technology</i>	2026
Best paper award from Korea Information Processing Society <i>A Study on Vulnerabilities and Defense Systems of ARM TrustZone-assisted TEEs</i>	2020
Best paper award from Korea Institute of Information Security and Cryptology <i>A Study on Isolation of Kernel Subsystems and Kernel Modules</i>	2020
Scholarship BK21+ Scholarship by the Ministry of Education of Korea	Mar 2020 - Aug 2025
Grants ACM CCS 2025 Student Travel Grant	Oct 2025

SERVICES & ACTIVITIES

IEEE S&P 2026 <i>Artifact Evaluation Committee</i>	
EuroSys 2025 <i>Shadow Program Committee</i>	
IEEE Transactions on Dependable and Secure Computing <i>Reviewer</i>	Oct 2025
Secondary-Reviewer	
IEEE Transactions on Computers	2025
Silicon Valley Cybersecurity Conference	2025

TALKS & SEMINARS

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- [1] "A Systematic Look at Mutual Privacy in AI Model Usage" presented at 한국정보보호학회 (KIISC) 차세대 저작권보호기술 워크숍, May 2026
 - [2] "Hardware-Assisted Security on Arm Mobile Platforms - From Memory Safety to Confidential Computing" presented at Pusan National University (online), December 2025
 - [3] "Hardware-Assisted Security on Arm Mobile Platforms - From Memory Safety to Confidential Computing" presented at Sejong University, November 2025

SKILLS

Programming: C/C++, Rust, Python

Frameworks: LLVM, rustc, ARM FVP, KVM/QEMU, SGX SDK

Platforms: Linux (x86_64, AArch64, AArch32), Android

Language: English (iBT TOEFL: 114), Korean (native)